Cour 03

Rhéologie (Viscosity of solutions)

- Viscosity is related to friction, which is a force that opposes the relative motion between two surfaces in contact.

- The frictional force is given by the following formula :

frictional force → $F = \eta \cdot A \cdot \frac{dv}{dx}$ ← velocity gradient

η : viscosity coefficient ; A : surface

$$\eta = \frac{F}{A} \cdot \frac{dx}{dv} = Kg.s^{-1}.m^{-1}$$



loi (Poiseuille) - Ostwald -



- Hopler - (loi de stokes)

Rotationelle (Sin - Cos)

- $\eta = \frac{N}{m^2} \cdot \frac{m}{\frac{m}{s}} = \frac{N}{m^2} \cdot s$

$= Kg.s^{-1}.m^{-1}$ (MKSA)

- η = pascal.s = poiseuille (PP)

$[\eta] = K.M^{\alpha}$

1 Poise = $1\ g.s^{-1}.cm^{-1}$ (CGS)

- Water at 25°c = $\eta_a = 10^{-3}$ Pa.s → خفيف في التدفق.
- Water at 100°c = $\eta = 0{,}28 \times 10^{-3}$ Pa.s → يصبح أخف.
- Glue at 25°c = $\eta = 10^{5}$ Pa.s → ثقيل (c° عامل حركي)
- Air at 20° c = $\eta = 1{,}8 \times 10^{-5}$ Pa.s → خفيف (les molecule sont libres).

Type of viscosity :

Dynamic viscosity :

$[\eta_{dym}] = Kg.m^{-1}.s^{-1} = Pa.s$

Kinematic viscosity :

$\eta_{cym} = \frac{\eta_{dym}}{\rho}$

in CGS :

1 stokes = $10^{-4}\ m^2.s^{-1}$

$[\eta_{cym}] = \frac{Kg.m^{-1}.s^{-1}}{Kg.m^{-3}} = m^2.s^{-1}$

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Classification of fluids :

$$\tau = \eta \cdot \frac{dv}{dx}$$

- τ : the shear stress.
- η : the viscosity of fluid
- $\frac{dv}{dx}$: the change of velocity with respect to distance

Factors influencing viscosity

- Nature of fluid
- Temperature
- Pressure
- Molecular weight
- C. of solute
- particles size and distribution.

The Reynolds number

$$R = \frac{\rho \cdot \bar{v} \cdot D}{\eta}$$



← laminar

← Turbulent.

Reynolds number :

- $Re < 2000 \Rightarrow$ laminar flow.
- $2000 < Re < 4000 \Rightarrow$ transitional flow.
- $Re > 4000 \Rightarrow$ Turbulent flow.

poiseuille law.

$$V(r) = \frac{\Delta P}{\Delta x} \cdot \frac{1}{4\eta} (R^2 - r^2)$$

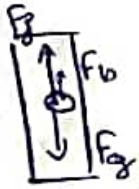
$t = 0 \quad V(0) = \frac{\Delta P}{\Delta x} \cdot \frac{1}{4\eta} \cdot R^2 = V_{max}$.

$$D = \frac{V}{t} = \frac{\Delta P}{\Delta x} \cdot \frac{\pi \cdot R^4}{8\eta} \quad \text{(flow Rate)} \ m^3/s$$

$$\begin{cases} \frac{V}{t} = \frac{\rho \cdot g \cdot h}{\Delta x} \cdot \frac{\pi \cdot R^4}{8\eta} \end{cases}$$

- la loi de stokes $\Rightarrow$ les masses volumiques + les forces.
- la loi de Poiseuille $\Rightarrow$ la pression
- une suspension : le soluté mélange pas totalement avec le solvan

stokes law



$$\sum F = 0$$

$$f_g = F_g - F_b$$

$$\begin{cases} 6\pi \cdot \eta \cdot r \cdot V = F_g \\ \frac{4}{3} \cdot \pi r^3 \rho_0 g = F_b \\ \frac{4}{3} \pi r^3 \cdot \rho \cdot g = F_g \end{cases}$$

$$V = \frac{2r^2 (\rho - \rho_0) g}{9\eta}.$$

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